Lab 2- GRADMAP PRODUCT SPECIFICATION

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CS 411W, spring 2026

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13 March 2026

Draft 1

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1. Introduction

GradMap is a web-based system that helps college students and academic advisors plan classes and keep track of degree progress. Many students have trouble graduating on time because degree requirements can be confusing, prerequisites can make scheduling difficult, and classes may conflict with each other. Advisors also spend a lot of time looking through student records and helping students figure out what courses they should take each semester.

GradMap helps make this process easier by putting degree requirements, course scheduling, and conflict detection all in one place. This allows students and advisors to see what classes are needed and plan future semesters more easily.

1.1 Purpose

Over the years universities have faced an increase in challenges when it comes to helping students graduate on time. Rising enrollment numbers and more complex degree requirements have made academic planning significantly more complicated. Many students are balancing full course loads as well as employment, internships, and families. These other obligations make manual degree tracking every semester challenging and error prone. Academic advisors face inefficiencies within existing systems. Many of these challenges come from outdated degree planning, inefficient course registration systems, as well as student obligation and conflicting scheduling . The purpose of the GradMap prototype is to demonstrate how an automated academic planning system can assist students and advisors in building conflict-free course schedules and tracking degree progress.

GradMap will allow students to enter their major, completed courses, and personal scheduling constraints such as work schedules or other obligations. The system will then generate possible semester schedules while verifying prerequisite requirements and detecting potential conflicts.

Academic advisors will be able to review student degree progress and assist with planning future semesters. By providing a centralized platform for scheduling and degree tracking, GradMap aims to reduce graduation delays and improve academic planning efficiency

1.2 Scope

The GradMap prototype will demonstrate the core functionality of the proposed system. The prototype will allow users to input degree information and generate semester schedules that satisfy prerequisite requirements and avoid time conflicts.

The system will include features for viewing degree progress, identifying missing prerequisite courses, and detecting scheduling conflicts. The prototype will simulate course scheduling and degree requirements to demonstrate the functionality of the system.

1.3 Definitions, Acronyms, and Abbreviations

Automated Smart Scheduling

A scheduling process where the system automatically builds personalized class plans based on each student's degree requirements, course availability, and preferences.

Conflict Detection

The system's ability to identify and alert users about scheduling overlaps or degree-requirement conflicts.

Conflict-Free Scheduling

A process that ensures no overlapping classes or time conflicts occur within a student's schedule.

Constraint Blocks

Reserved time periods (for work, personal, or other obligations) when classes cannot be scheduled.

Continuous Integration and Continuous Deployment (CI/CD)

Automated workflows that build, test, and deploy GradMap updates using GitHub Actions.

Database (MySQL)

Stores user data, course information, degree requirements, and scheduling records.

Degree Requirements

The specific set of courses and academic conditions a student must complete to earn a degree.

Graphical User Interface (GUI)

The visual component of GradMap that users interact with — menus, buttons, and forms.

Integrated Development Environment (IDE)

A software tool — Visual Studio Code — used by developers to write and test GradMap's code.

Load-Balancing Recommendations

A GradMap feature that suggests optimal course combinations to balance workloads and prevent burnout.

Server-Side Language (Java)

The programming language used on the back end of GradMap to handle data processing and logic.

Streamlined Workflow

A simplified, guided scheduling process that helps users navigate course planning efficiently.

Version Control (Git / GitHub)

Tracks and manages changes in GradMap's codebase, allowing collaboration among developers.

Web Server (Apache, Nginx, or IIS)

Hosts and delivers the GradMap web application to users' browsers.

1.4 References

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1.5 Overview

This product specification describes GradMap's prototype and its designs. It will explain the hardware and software, the systems architecture, prototype features, and interfaces that will be used to support the functionality of the system.

2. Overall Description

GradMap is a web based application meant for scheduling and degree planning to assist college students and academic advisors in creating conflict free course schedules and structured degree plans. Universities increasingly face challenges related to overlapping class times and conflicts between academic requirements and personal commitments which can contribute to delayed graduation. The system addresses common challenges such as overlapping class times, unmet prerequisites, and conflicts between academic and personal commitments that frequently delay graduation by integrating degree requirements, real time course availability, student specific constraints, and pre and co reqs all into one system. The primary objective of GradMap is to streamline the scheduling process by integrating degree requirements, course availability, and student constraints into a single platform. By automating schedule generation while preserving advisor oversight, GradMap reduces manual workload, minimizes scheduling errors, and supports on time graduation.

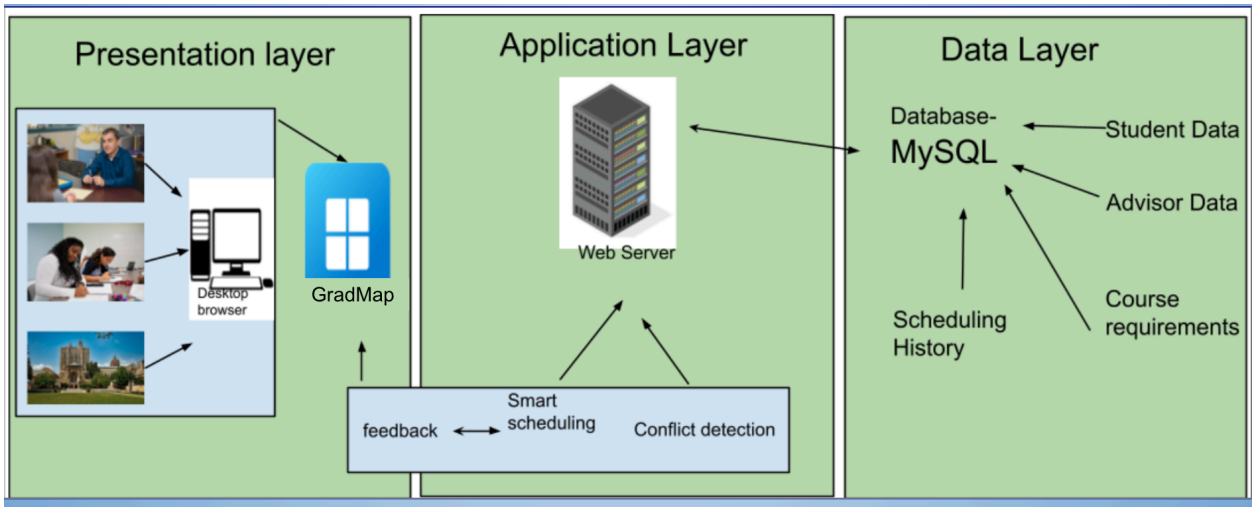
2.1 Product Perspective

GradMap will function as a web application that interacts with university course data and degree requirements to assist students in planning their academic schedules. The system is designed to complement existing university systems rather than replace them.

The prototype will include a graphical user interface that allows students and advisors to input academic information, review degree progress, and generate potential semester schedules. The system may interact with external data sources or services through APIs when retrieving course information or scheduling data.

GradMap will follow a layered architecture consisting of a user interface layer, an application logic layer, and a data storage layer.

Figure 1- GradMap Prototype Major Functional Component Diagram



2.2 Prototype Functions

The GradMap prototype will demonstrate several core system features.

Students will be able to enter their major and completed courses. The system will analyze this information and determine which courses remain for degree completion. Students will also be able to enter scheduling constraints such as work hours or other obligations.

GradMap will generate possible semester schedules that satisfy prerequisite requirements and avoid time conflicts. The system will also display degree progress so students can see which requirements have been completed and which courses are still needed.

Academic advisors will be able to review student progress and verify that generated schedules follow the correct course sequence.

2.3 User Characteristics

N/A

2.4 Constraints

N/A

2.5 Assumptions and dependencies

N/A